

## ALENCON-VALFRAMBERT AP, France

WMO: 071390

Lat: 48.4456N

Lon: 0.1103E

Elev: 141

StdP: 99.64

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -4.9       | -3.2 | -8.0                          | 1.9 | -2.7 | -5.9 | 2.3 | -2.2 | 11.5                  | 9.4  | 10.4 | 9.4  | 2.2                   | 30   | 0.542 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.6                   | 30.0            | 19.6 | 27.7 | 18.8 | 25.8 | 18.2 | 20.7                | 27.6 | 19.7 | 25.8 | 18.9 | 24.3 | 3.3                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.4                            | 13.5 | 23.1 | 17.5 | 12.8 | 22.1 | 16.7 | 12.1 | 21.1 | 60.2          | 27.9 | 56.9 | 26.0 | 54.0 | 24.4 | 24.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |  |
| 9.3               | 8.2  | 7.3 | DB                         | -7.8 | 33.4               | 2.8 | 2.5                                                | -9.9  | 35.2       | -11.5 | 36.7       | -13.1 | 38.1       | -15.2 | 40.0 |  |
|                   |      |     | WB                         | -8.2 | 22.4               | 2.8 | 1.2                                                | -10.2 | 23.3       | -11.8 | 24.0       | -13.4 | 24.7       | -15.4 | 25.6 |  |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 11.2        | 4.5   | 5.1   | 7.4   | 9.9   | 13.2  | 16.6  | 18.6  | 18.3  | 15.2  | 12.0  | 7.7   | 4.9   |
|                                                                   | DBStd       | 6.00        | 3.93  | 3.51  | 3.16  | 3.18  | 3.10  | 3.07  | 2.95  | 2.89  | 2.74  | 3.19  | 3.34  | 3.82  |
|                                                                   | HDD10.0     | 708         | 172   | 138   | 90    | 41    | 7     | 0     | 0     | 0     | 1     | 17    | 83    | 160   |
|                                                                   | HDD18.3     | 2725        | 429   | 370   | 338   | 254   | 162   | 71    | 32    | 36    | 102   | 198   | 319   | 415   |
|                                                                   | CDD10.0     | 1128        | 2     | 1     | 11    | 36    | 106   | 197   | 268   | 258   | 155   | 78    | 13    | 3     |
|                                                                   | CDD18.3     | 103         | 0     | 0     | 0     | 0     | 2     | 17    | 42    | 36    | 6     | 0     | 0     | 0     |
|                                                                   | CDH23.3     | 1094        | 0     | 0     | 0     | 6     | 37    | 197   | 412   | 357   | 82    | 4     | 0     | 0     |
|                                                                   | CDH26.7     | 324         | 0     | 0     | 0     | 0     | 3     | 56    | 126   | 122   | 17    | 1     | 0     | 0     |
| Wind                                                              | WSAvg       | 3.4         | 3.7   | 3.8   | 3.7   | 3.5   | 3.4   | 3.2   | 3.2   | 3.0   | 3.0   | 3.0   | 3.3   | 3.6   |
| Precipitation                                                     | PrecAvg     | 750         | 76    | 61    | 53    | 54    | 61    | 52    | 55    | 52    | 56    | 73    | 74    | 83    |
|                                                                   | PrecMax     | 1021        | 160   | 135   | 160   | 130   | 100   | 131   | 141   | 131   | 142   | 173   | 157   | 186   |
|                                                                   | PrecMin     | 532         | 11    | 8     | 6     | 9     | 10    | 4     | 7     | 5     | 13    | 23    | 24    | 21    |
|                                                                   | PrecStd     | 148         | 37    | 36    | 34    | 34    | 27    | 30    | 35    | 35    | 38    | 37    | 33    | 38    |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 13.1  | 14.8  | 19.6  | 24.1  | 26.7  | 31.5  | 32.8  | 33.8  | 28.9  | 23.1  | 16.7  | 13.7  |
|                                                                   |             | MCWB        | 11.8  | 10.5  | 12.4  | 15.3  | 18.4  | 21.3  | 20.7  | 21.0  | 18.6  | 16.9  | 14.5  | 12.4  |
|                                                                   | 2%          | DB          | 12.1  | 12.4  | 16.8  | 21.2  | 24.2  | 28.0  | 29.7  | 29.9  | 25.8  | 20.3  | 14.8  | 12.6  |
|                                                                   |             | MCWB        | 11.0  | 10.0  | 11.4  | 14.2  | 16.9  | 19.3  | 19.4  | 19.6  | 17.4  | 15.9  | 13.0  | 11.5  |
|                                                                   | 5%          | DB          | 11.0  | 11.2  | 14.5  | 18.4  | 21.9  | 25.4  | 27.7  | 27.1  | 23.1  | 18.2  | 13.7  | 11.6  |
|                                                                   |             | MCWB        | 9.8   | 9.5   | 10.8  | 12.7  | 15.8  | 18.2  | 18.8  | 18.6  | 16.3  | 14.8  | 12.1  | 10.6  |
|                                                                   | 10%         | DB          | 9.8   | 10.2  | 12.7  | 15.9  | 19.6  | 23.2  | 25.4  | 24.7  | 20.9  | 16.8  | 12.6  | 10.6  |
|                                                                   |             | MCWB        | 8.7   | 8.7   | 10.0  | 11.4  | 14.5  | 17.1  | 18.2  | 17.8  | 15.6  | 14.4  | 11.2  | 9.6   |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 12.1  | 11.6  | 13.5  | 16.7  | 19.3  | 22.2  | 21.7  | 22.1  | 19.9  | 18.1  | 15.0  | 12.8  |
|                                                                   |             | MCDB        | 12.9  | 13.2  | 17.1  | 22.4  | 25.1  | 29.3  | 29.8  | 31.1  | 26.2  | 21.5  | 15.8  | 13.3  |
|                                                                   | 2%          | WB          | 11.2  | 10.6  | 12.3  | 14.6  | 17.8  | 20.2  | 20.5  | 20.6  | 18.5  | 16.6  | 13.4  | 11.7  |
|                                                                   |             | MCDB        | 12.0  | 11.9  | 15.1  | 19.7  | 23.0  | 26.0  | 27.6  | 27.5  | 23.6  | 19.1  | 14.6  | 12.4  |
|                                                                   | 5%          | WB          | 10.0  | 9.7   | 11.4  | 13.2  | 16.3  | 18.9  | 19.6  | 19.5  | 17.3  | 15.7  | 12.3  | 10.7  |
|                                                                   |             | MCDB        | 10.8  | 11.0  | 13.7  | 17.4  | 20.6  | 24.3  | 25.9  | 25.4  | 21.8  | 17.6  | 13.4  | 11.5  |
|                                                                   | 10%         | WB          | 8.9   | 8.8   | 10.4  | 11.9  | 15.0  | 17.7  | 18.7  | 18.4  | 16.3  | 14.7  | 11.4  | 9.7   |
|                                                                   |             | MCDB        | 9.7   | 10.0  | 12.6  | 15.3  | 18.9  | 22.2  | 24.1  | 23.4  | 19.7  | 16.4  | 12.4  | 10.5  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.2   | 6.5   | 8.4   | 9.8   | 9.6   | 10.2  | 10.6  | 10.8  | 10.3  | 8.1   | 6.0   | 5.3   |
|                                                                   |             | MCDBR       | 5.6   | 8.0   | 11.8  | 14.2  | 13.4  | 14.3  | 14.7  | 15.3  | 14.4  | 10.0  | 6.9   | 5.3   |
|                                                                   | 5% WB       | MCWBR       | 4.9   | 6.0   | 7.3   | 8.0   | 7.3   | 6.9   | 6.1   | 6.4   | 7.1   | 6.1   | 5.3   | 4.8   |
|                                                                   |             | MCWBR       | 5.3   | 6.9   | 9.7   | 12.5  | 12.1  | 12.6  | 12.8  | 13.0  | 11.9  | 8.8   | 6.2   | 5.2   |
| Clear-Sky Solar Irradiance                                        | taub        |             | 0.321 | 0.340 | 0.371 | 0.393 | 0.399 | 0.402 | 0.396 | 0.387 | 0.366 | 0.356 | 0.330 | 0.315 |
|                                                                   |             | taud        | 2.428 | 2.382 | 2.290 | 2.234 | 2.261 | 2.289 | 2.304 | 2.350 | 2.389 | 2.418 | 2.434 | 2.402 |
|                                                                   | Ebn at Noon |             | 743   | 810   | 837   | 854   | 863   | 861   | 859   | 852   | 835   | 779   | 723   | 695   |
|                                                                   |             | Edh at Noon | 66    | 86    | 110   | 128   | 130   | 127   | 124   | 113   | 99    | 82    | 65    | 60    |
| All-Sky Solar Radiation                                           | RadAvg      |             | 0.97  | 1.76  | 2.88  | 4.36  | 5.14  | 5.73  | 5.46  | 4.75  | 3.72  | 2.16  | 1.18  | 0.82  |
|                                                                   | RadStd      |             | 0.10  | 0.22  | 0.36  | 0.49  | 0.45  | 0.59  | 0.40  | 0.43  | 0.28  | 0.18  | 0.12  | 0.11  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (36 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page